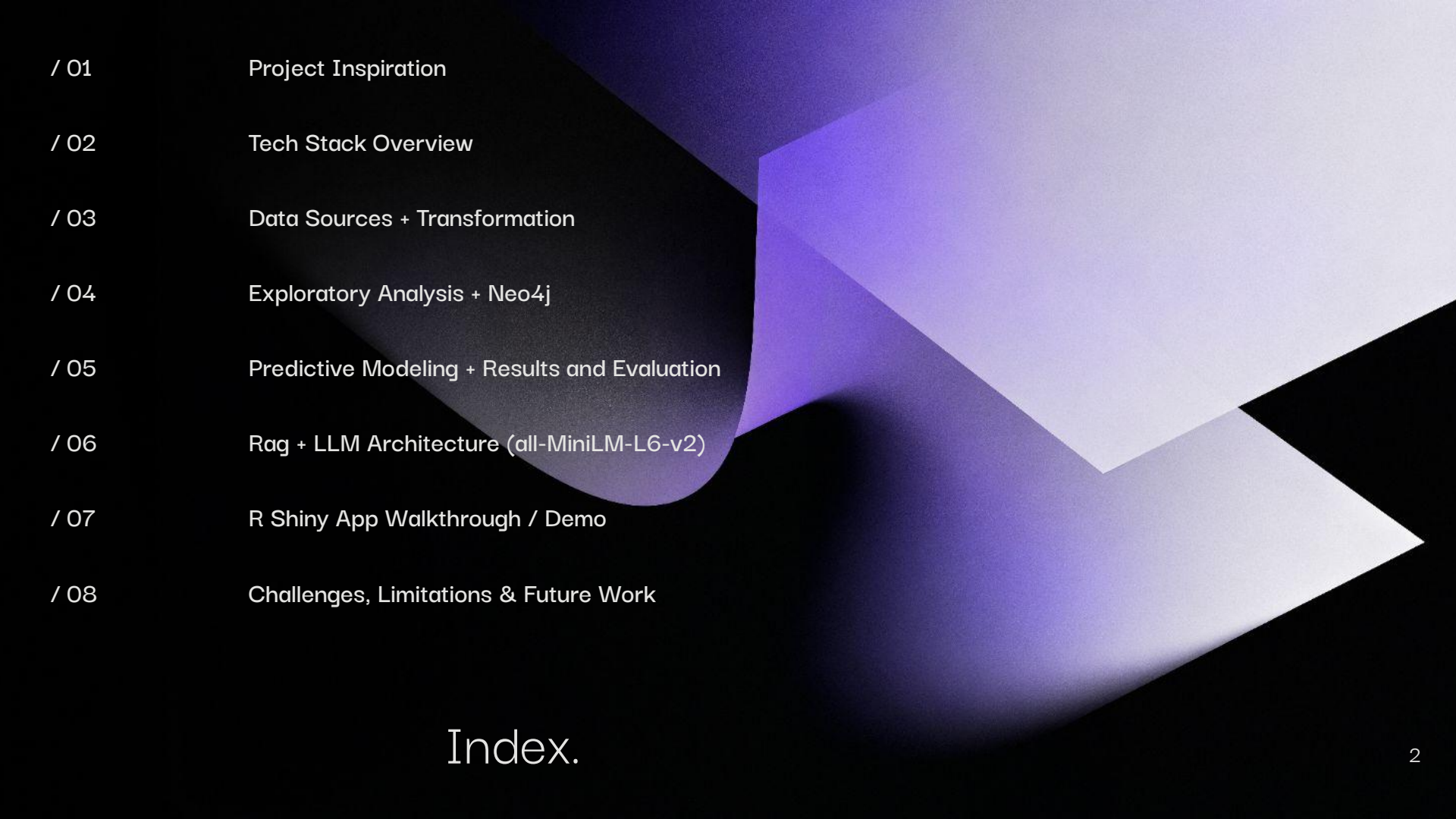


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NLP Email Optimization RAG + OpenAI + Mailchimp

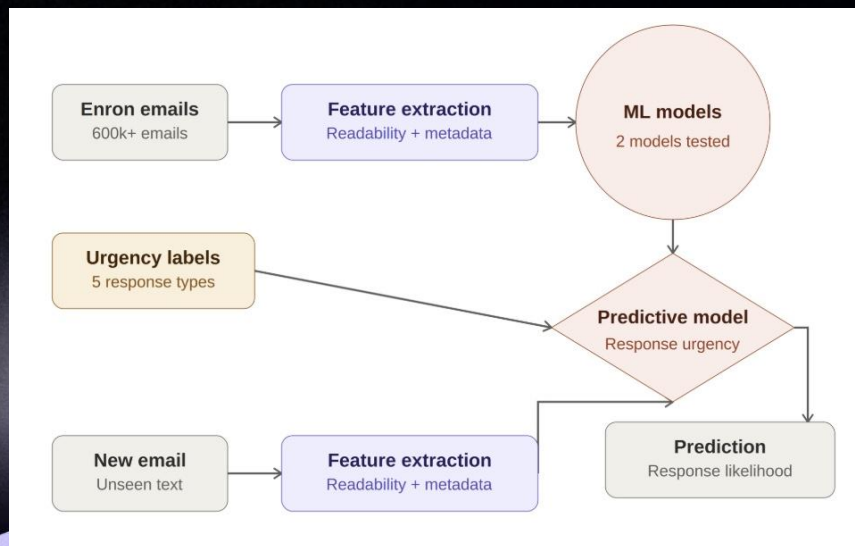
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 - / 05 Predictive Modeling + Results and Evaluation
 - / 06 Rag + LLM Architecture (all-MiniLM-L6-v2)
 - / 07 R Shiny App Walkthrough / Demo
 - / 08 Challenges, Limitations & Future Work

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/ 01

Project Inspiration.

Overview & insights.



The idea for this project all began when I discovered natural language processing and text classification. I realized that in this future age of technology, human correspondence is quietly disappearing. I wanted to understand how AI is processing emails at scale and which words make an impact on response rate.

https://mediacenter.ibm.com/media/Text+Classification+AI+Techniques+and+Real-World+Applications/1_o3chonwn

Emails are no longer being processed by humans, but by machines that categorize text based on keywords. Certain morphemes signify urgency and automated systems are processing them at scale.

The goal of this project is to create a tool that predicts the likelihood of an email response based on text features alone. As a stretch goal I wanted to Infuse an LLM with real world marketing savvy to increase consumer impact.

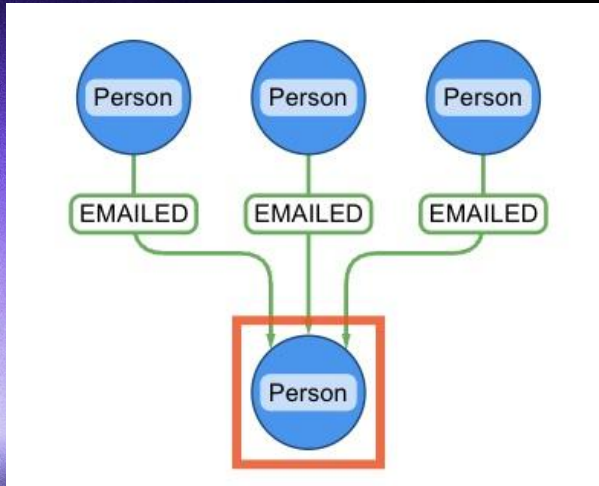
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Tech Stack



Technology and Libraries accessed.

<https://www.yworks.com/yed-live/>



/ 01 UI framework is compiled with R Shiny, bslib and plotly. Network Analysis Visualized in a Jupyter notebook using Neo4j (GraphWidget and GraphDatabase)

/ 02 ML and Backend development consists of R and Python code. Libraries include reticulate, scikit-learn, pandas, numpy, matplotlib and syllables.

/ 03 AI and Rag layer is composed of fastembed (all-MiniLM-L6-v2), OpenRouter (gpt-oss-20b), OpenAI Python client and Mailchimp knowledge base.

Data comes from Kaggle Enron email-reply dataset, Enron mass email corpus and 25 curated best practice email chunks available at Mailchimp.com

Note: Heatmap, ROC curve and all other visualizations are built with Matplotlib and/or Seaborn.

/ 03

Data Sources + Transformation

Enron Email-Reply Dataset.

<https://www.kaggle.com/datasets/oanannv/enron-email-reply-dataset>



/ 01 When Enron collapsed in 2001, federal investigators released over 600,000 internal company emails as public record. It is the largest public-domain email corpus that exists

/ 02 I chose this dataset so that I could model email response time. There are 400,000 real corporate emails that I use to uncover network structure, response latency, and hierarchy in this project

/ 03 Between a clean Kaggle dataset and the original messy 600k email dump I was able to merge them and add features like response time, a response indicator and various text features to predict response time.

Text Features include 'char_count', 'exclamation_count', 'token_count', 'avg_word_length', 'question_mark', 'stop_word_count', 'syllable_count'

Mailchimp Knowledge Base & Best Practice.

<https://mailchimp.com/help/best-practices-mailchimp-email/>



The Mailchimp knowledge base is 25 email marketing best-practice tips covering subject lines, tone, urgency and send times these are embedded with fastembing. When a user submits an email, the system finds the 5 most relevant tips for that specific email and sends them to the LLM along with the email's predicted response score, so the AI rewrites the email using real, research-backed guidance instead of generic suggestions. The Rag system uses the response score infused with knowledge from Mailchimp to help prompt the **openai/gpt-oss-20b:free model**.

- Subject Lines
- Email Length and Structure
- Tone and Clarity
- Urgency and Engagement
- Mobile and Formatting
- Sending Time

/ 04

Exploratory Analysis + Neo4j

Exploratory Analysis.

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god. Despite being closer to the Sun than Venus, its temperatures aren't as terribly hot as that planet's. Its surface is quite similar to that of Earth's Moon, which means there are a lot of craters and plains. Speaking of craters, many of them were named after artists or authors who made significant contributions to their respective fields. Mercury takes a little more than 58 days to complete its rotation, so try to imagine how long days must be there!

Transforming processes with precision.



Fragmented user journey

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god.

/ BEFORE



Seamless user journey

Venus has a beautiful name and is a very hot planet. It's the second planet from the Sun. Its atmosphere is extremely poisonous, primarily made up of carbon dioxide with clouds of sulfuric acid.

/ AFTER

Anticipating risks & challenges.

/ 01

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System.

/ 02

Mars is full of iron oxide dust, which gives the planet its reddish cast. It's made of basalt.

/ 03

- bslib
- plotly

R and Python for the Backend

- reticulate
- scikit-learn
- pandas / numpy
- syllables

I used reticulate to utilize Python models in the R environment.

Jupyter notebook

- data cleaning

“Venus is the second planet from the Sun and is extremely hot. It has a beautiful name”

Overview & insights.

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god. Despite being closer to the Sun than Venus, its temperatures aren't as terribly hot as that planet's. Its surface is quite similar to that of Earth's Moon, which means there are a lot of craters and plains. Speaking of craters, many of them were named after artists or authors who made significant contributions to their respective fields. Mercury takes a little more than 58 days to complete its rotation, so try to imagine how long days must be there!

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precision, evolve
with flow.

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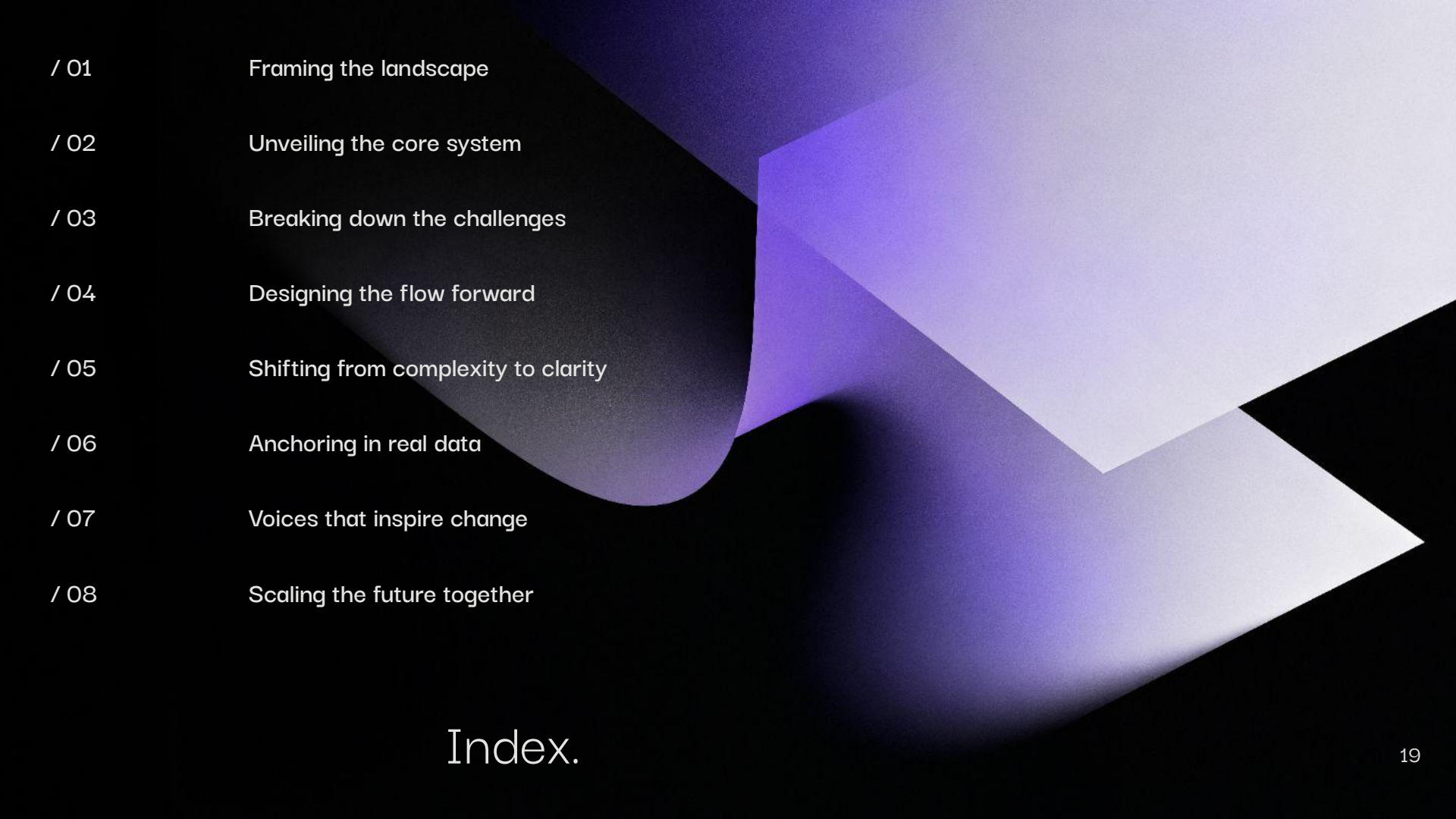
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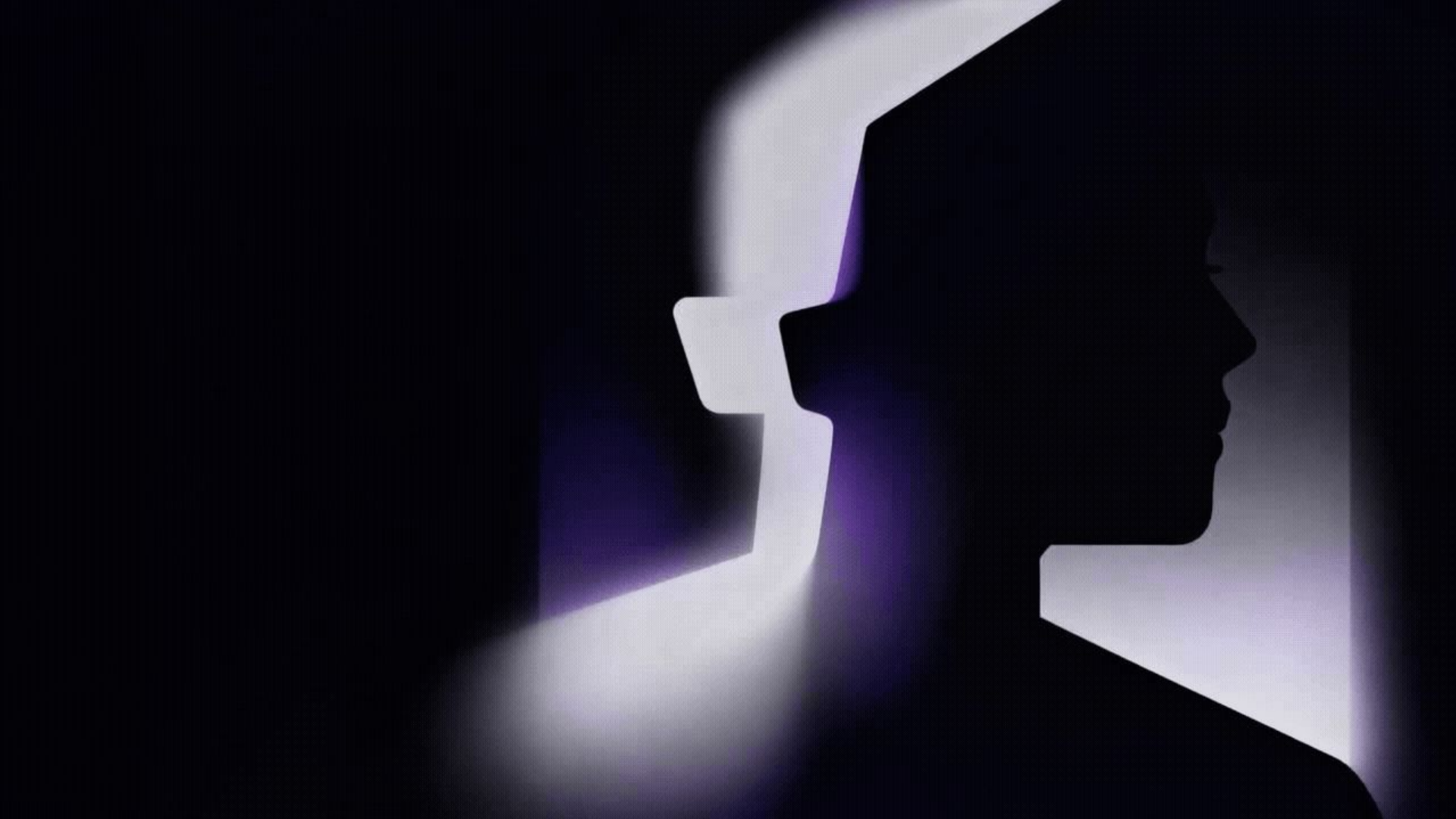
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- / 01 Framing the landscape
 - / 02 Unveiling the core system
 - / 03 Breaking down the challenges
 - / 04 Designing the flow forward
 - / 05 Shifting from complexity to clarity
 - / 06 Anchoring in real data
 - / 07 Voices that inspire change
 - / 08 Scaling the future together

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Anticipating risks & challenges.

/ 01 Mercury is the closest planet to the Sun and the smallest one in the entire Solar System.

/ 02 Mars is full of iron oxide dust, which gives the planet its reddish cast. It's made of basalt.

/ 03 Earth is the third planet from the Sun and the only one known to harbor human life in the Solar System.

“Venus is the second planet from the Sun and is extremely hot. It has a beautiful name”

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“Venus is the second planet from the Sun and is extremely hot. It has a beautiful name”

Earth is the third planet from the Sun and the only one known that harbors human life.

/ 20xx

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System.

/ 20xx

Jupiter is a huge gas giant and the biggest planet in the entire Solar System.

/ 20xx

Venus has a beautiful name and it's the second planet in our Solar System.

/ 20xx

Designing the flow forward.

Transforming processes with precision.



Fragmented user journey

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god.

/ BEFORE



Seamless user journey

Venus has a beautiful name and is a very hot planet. It's the second planet from the Sun. Its atmosphere is extremely poisonous, primarily made up of carbon dioxide with clouds of sulfuric acid.

/ AFTER

“This is a quote filled with insight, meant to inspire and move forward: Mercury is the closest planet to the Sun and the smallest one in the entire Solar System”

—Jenna Doe.



“This is a quote filled with insight, meant to inspire and move forward: Mercury is the closest planet to the Sun and the smallest one in the entire Solar System”

—Jenna Doe.

The numbers behind the impact.

12

The number of Earth years it takes for Jupiter to complete one full orbit around the Sun

8

Planets in the Solar System

9.8

Earth's gravity in m/s^2

3.7

Mars gravity in m/s^2

24

Hours in an Earth day

“Explore the fundamental metrics shaping our understanding of the solar system”

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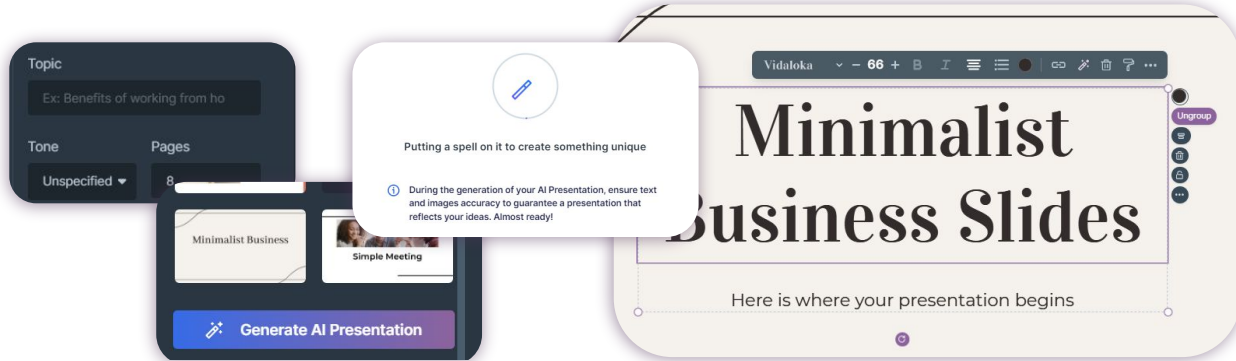
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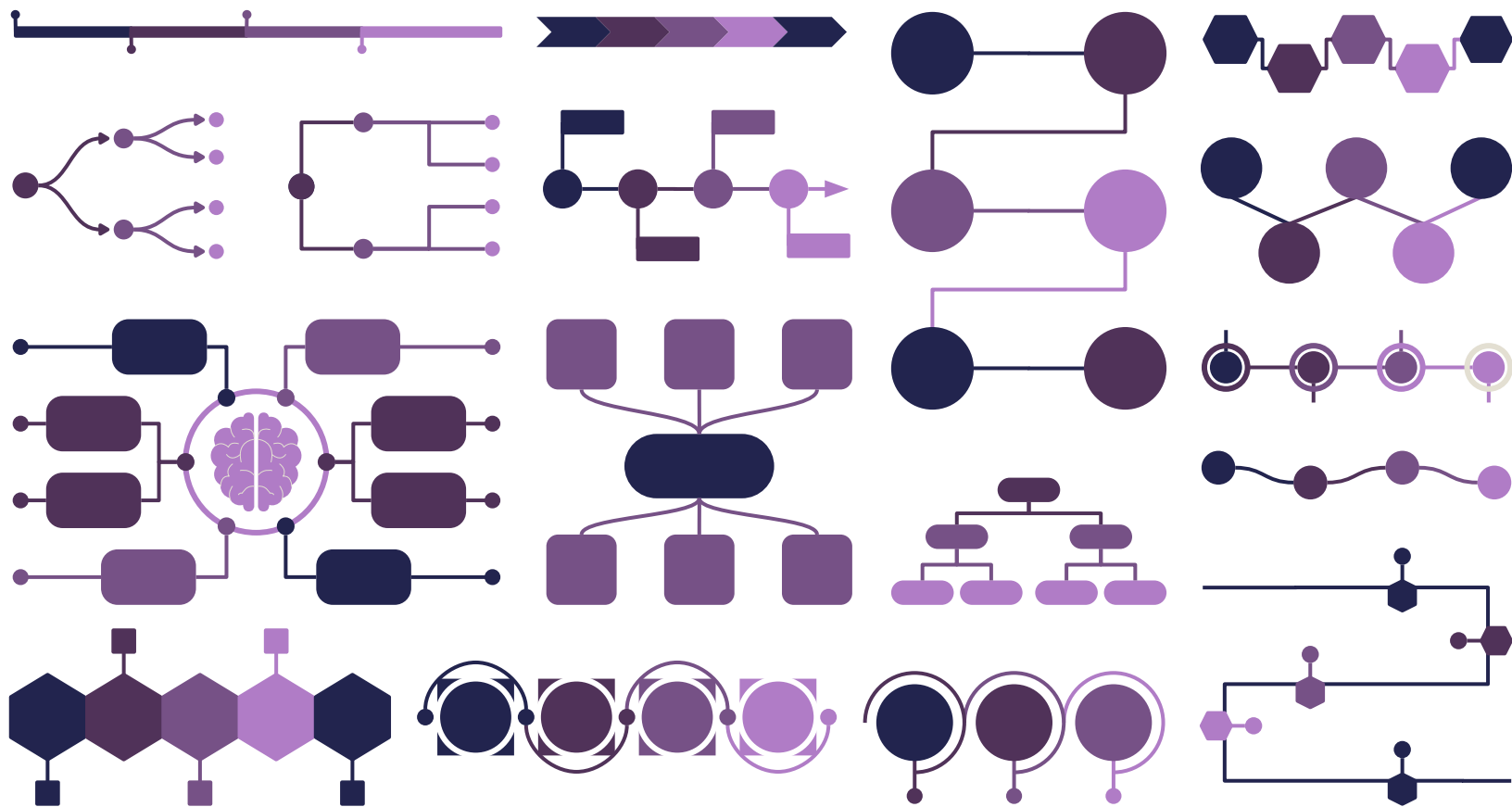
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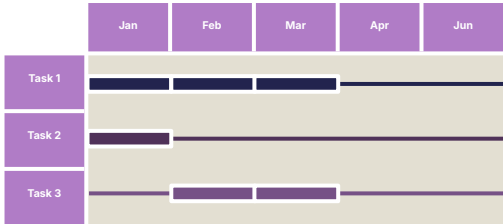
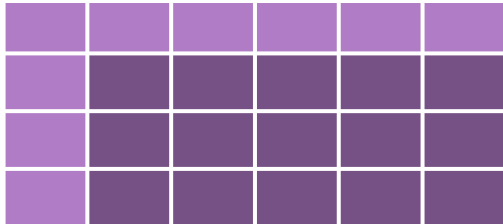
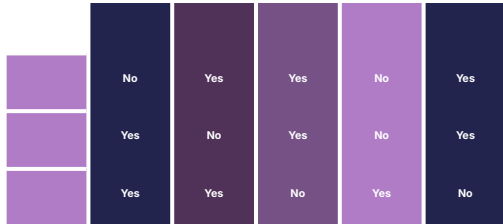
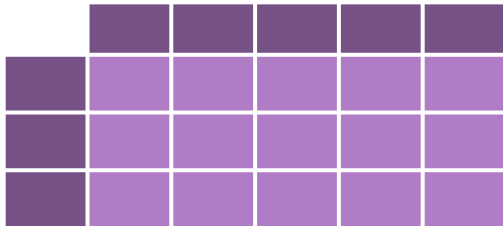
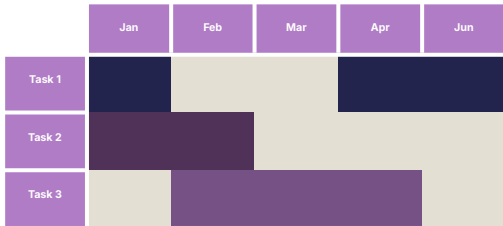
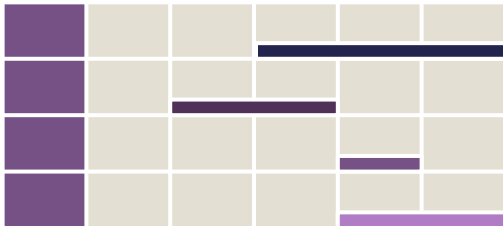
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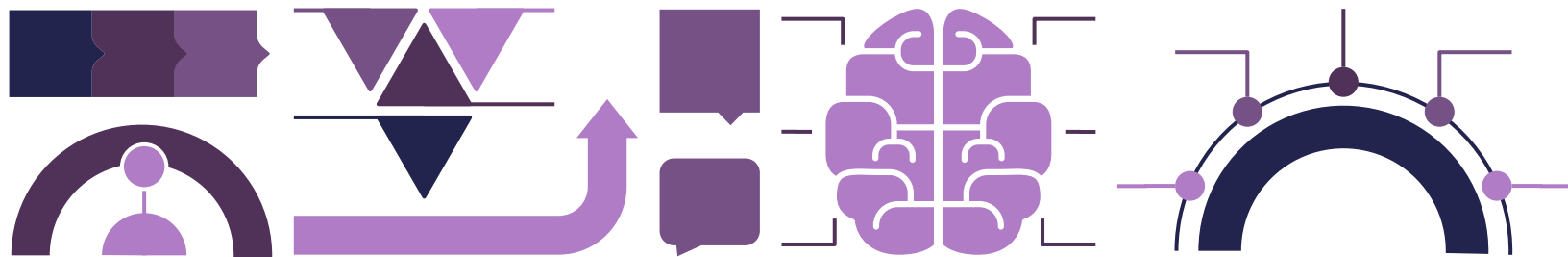
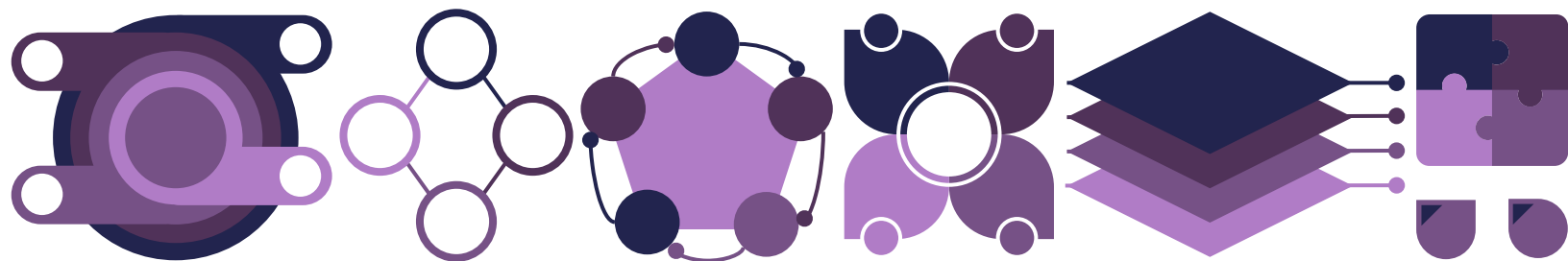
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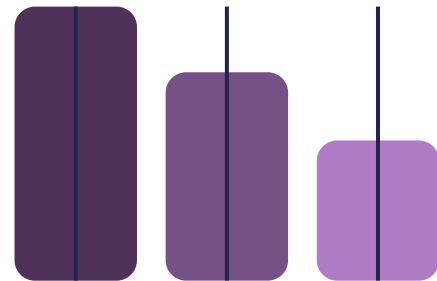
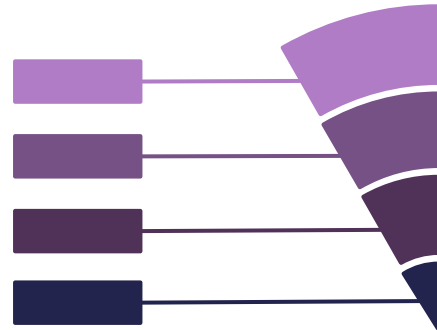
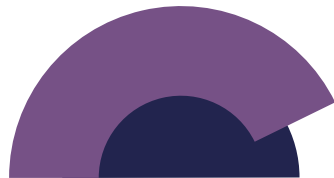
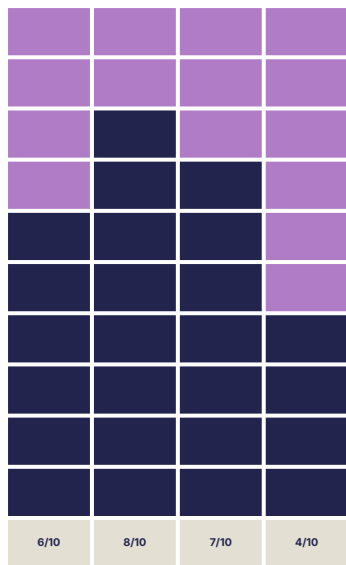












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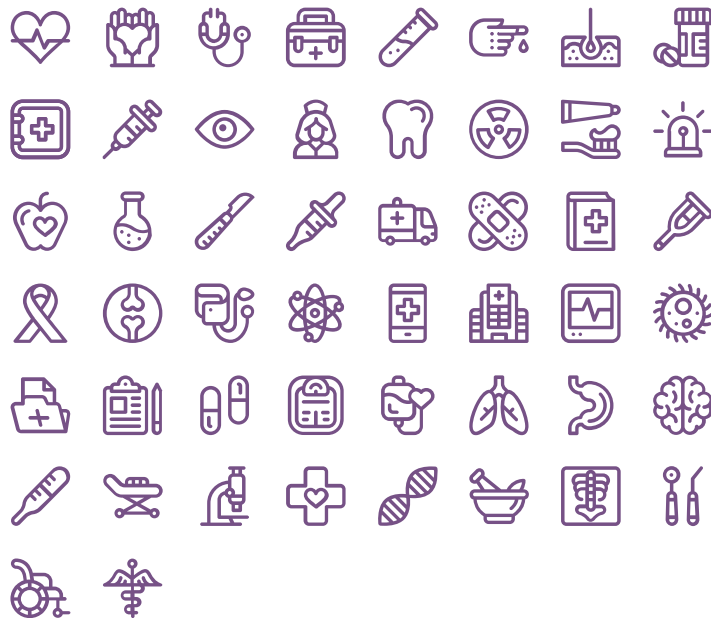
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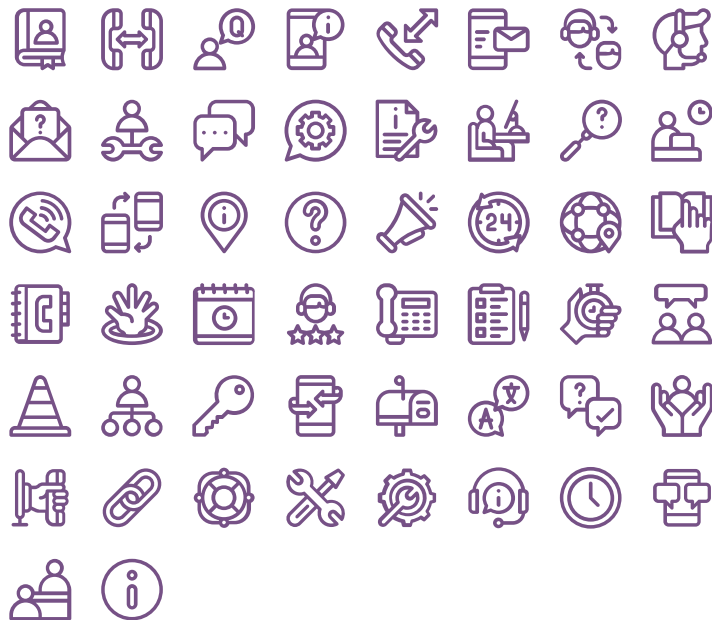
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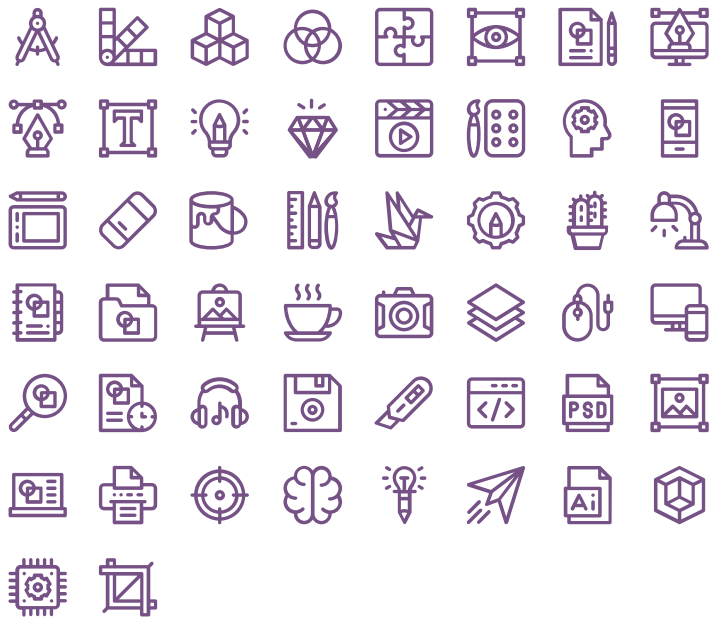
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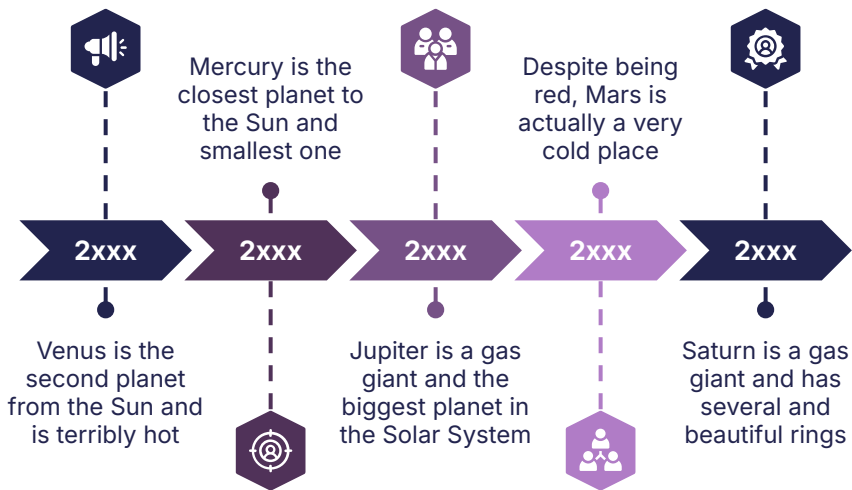
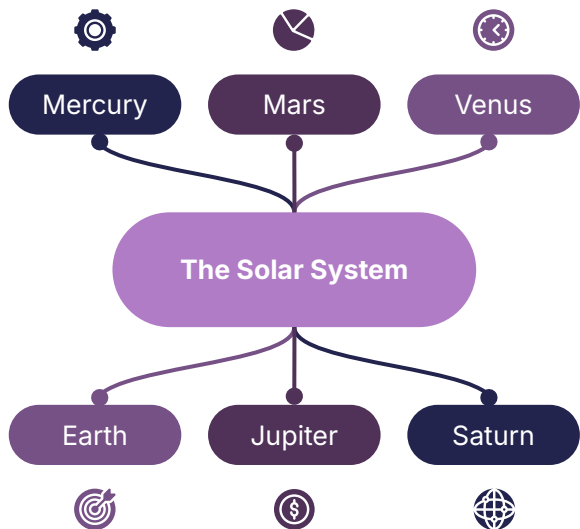


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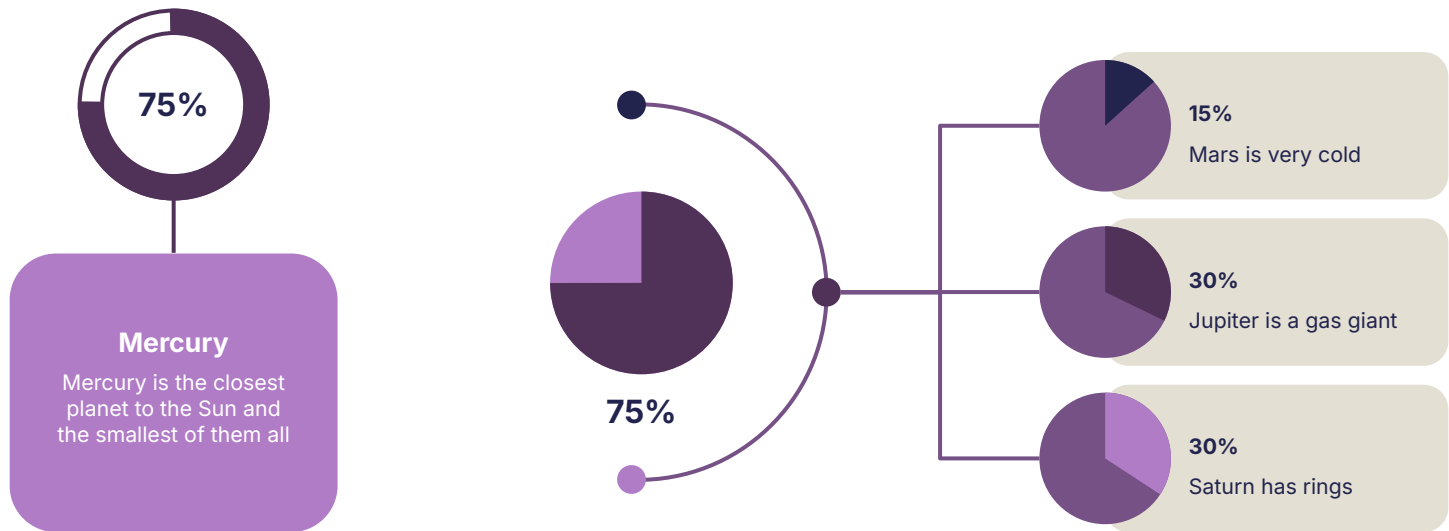


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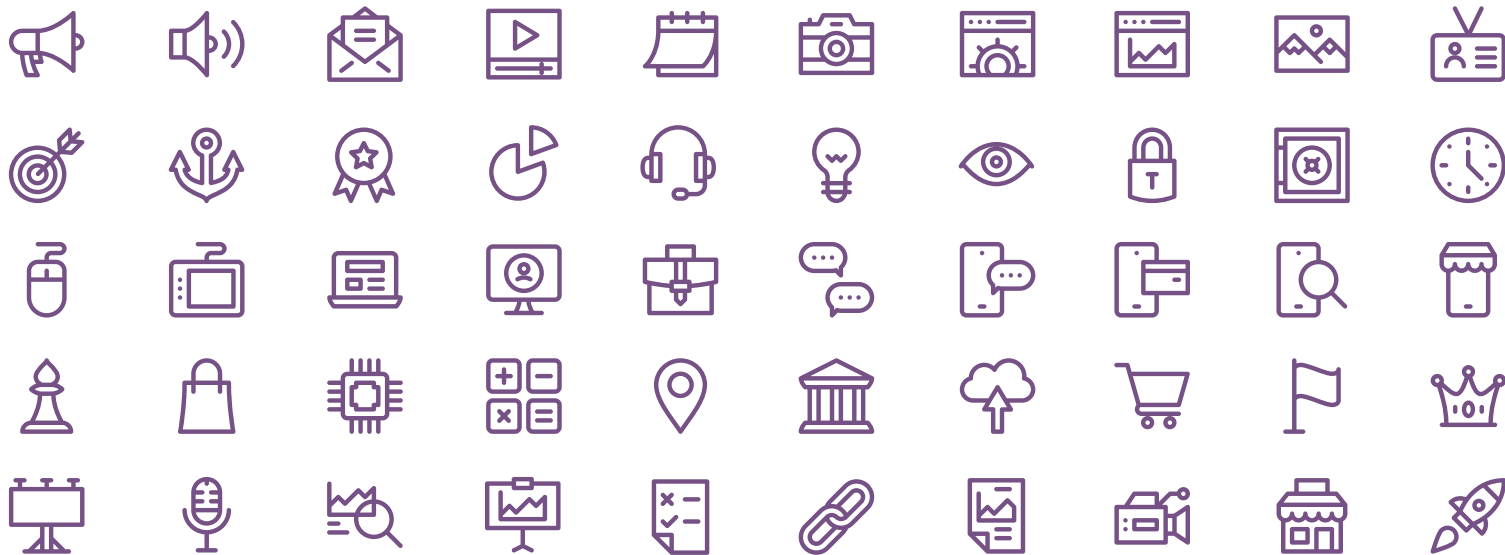
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